Computational Supplement

Frame covariance of CMB lensing in the exact $1+3$ covariant radiation hierarchy

Tim Gebbie

v1.0.0 — First public analytical reproducibility release
5 October 2026

1 Scientific scope

This supplement records the analytical companion to *Frame covariance of CMB lensing in the exact* $1+3$ *covariant radiation hierarchy*. The bundle is an independent Python reconstruction, from first $1+3$ covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000), `doi:10.1006/aphy.2000.6034`. It is archived at `doi:10.25375/uct.34069509`, active on publication of v1.0.0.

It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code, and it inherits no other code's conventions. It reproduces; it does not argue. No new effect is claimed.

The background expansion is *derived* from the $1+3$ covariant energy constraint rather than transcribed from a printed expansion law, so that the antecedent's own Eqs. (185) and (G.3) are checks on this bundle and not its sources. That ordering is what makes a reconstruction an audit, and it is load bearing: a port would have inherited the misprint recorded below.[1]

Department of Statistical Sciences, University of Cape Town. `tim.gebbie@uct.ac.za`

2 Theory-to-computation map

Table 1 links each published equation the numerics implement to its provenance and to the `.py` file and object implementing it, together with the test that checks it. Table 2 gives the notation and the names the quantities carry in the code.

Both are *generated* from `config/implementation-register.toml` by `scripts/make_tables.py`, and `tests/test_register.py` resolves every module, object and test named in them. They cannot drift from the code: a rename breaks the build rather than quietly falsifying a table printed here.

The harmonic normalisation is $Q_{A_\ell} = (-k_{\rm phys})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.[2]

Nomenclature is kept strictly separate: τ_ℓ are **mode** coefficients and τ_{A_ℓ} are **multipole** coefficients. *Annals II* notes that the distinction is not made in Bardeen-variable treatments. Every released figure plots a multipole mean-square, which holds for general geometries, rather than a mode mean-square, which holds only for almost-Robertson–Walker ones.

[1] The Gauss constraint at (G.3) prints $+\frac{2}{3}\Theta^2$ where the companion thesis and the Cargèse lectures Eq. (55) carry $-\frac{2}{3}\Theta^2$. In the Friedmann limit the printed sign gives $H^2 = K/a^2 - (\mu+\Lambda)/3$, which for a flat section is $-(\mu+\Lambda)/3$ and so has no solution at positive energy density. A misprint, recorded; the derived background already carries the correct sign, and a test now pins it so that a later edit towards the printed form fails rather than passes.

[2] *Annals I* prints $(-1)^\ell$ at the corresponding step where its own definition gives i^ℓ. The covariant multipole τ_{A_ℓ} is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests. A misprint, recorded in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md`.

Table 1: Equations implemented in the numerics, with their provenance and the code that implements them. Equation numbers are those of the *published* papers. Rows with no code are not yet implemented.

Quantity	**Source**	**Equation**	**File**	**Object**	**Checked by**
$\beta_\ell = O^{A_\ell} O_{A_\ell} = (\ell!)^2 2^\ell/(2\ell)!$	Annals I	(24)	`functions/harmonics/weights.py`	`beta`	`test_beta_module_matches_symbolic_exactly`
$\beta_\ell = \frac{\ell}{2\ell-1}\beta_{\ell-1}$, $\beta_{\ell+1} = \frac{\ell+1}{2\ell+1}\beta_\ell$	Annals I	(25)	`functions/harmonics/weights.py`	`beta`	`test_beta_recurrence_descending_general_ell`
$\Delta_\ell = 4\pi\beta_\ell/(2\ell+1)$	Annals I	(119)	`functions/harmonics/weights.py`	`delta_over_four_pi`	—
α_ℓ, defined by $\beta_\ell = \alpha_\ell^{-1}(2\ell+1)$	Annals II	App. F, p. 380	`functions/harmonics/weights.py`	`alpha`	`test_f4_is_direct_substitution_into_f3`
free-streaming weight $w_\ell = (\ell+1)^2/[(2\ell+3)(2\ell+1)]$	Annals II	(F.1), p. 379	`functions/harmonics/weights.py`	`free_streaming_weight`	`test_f1_weight_is_the_mode_recursion_weight_shifted`
covariant recursion $-\dot{\tau}_\ell = (k_{\rm com}/a)[w_\ell\tau_{\ell+1} - \tau_{\ell-1}]$	Annals II	(F.1), p. 379	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
β-normalised recursion, conformal time	Annals II	(F.3), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_the_three_normalisations_are_one_solution`
α-normalised recursion	Annals II	(F.4), p. 380	`functions/harmonics/free_streaming.py`	`free_stream`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
free-streaming solution $\alpha_\ell^{-1}\tau_\ell = j_\ell(k_{\rm com}\Delta\eta)$	Annals II	App. F, p. 380	`functions/harmonics/free_streaming.py`	`analytic_projection`	`test_f4_variable_is_exactly_the_spherical_bessel_function`
$\dot{\Theta}_\ell = k[\frac{\ell}{2\ell-1}\Theta_{\ell-1} - \frac{\ell+1}{2\ell+3}\Theta_{\ell+1}]$	Hu & Sugiyama	(6), p. 2601	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{F}_{\nu\ell} = \frac{k}{2\ell+1}[\ell F_{\nu(\ell-1)} - (\ell+1)F_{\nu(\ell+1)}]$	Ma & Bertschinger	(49), (50)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`
$\dot{\Delta}_{T\ell} = \frac{k}{2\ell+1}[\ell\Delta_{T(\ell-1)} - (\ell+1)\Delta_{T(\ell+1)}]$	Seljak & Zaldarriaga	(3d)	`functions/harmonics/external.py`	`WEIGHTS`	`test_external_hierarchies_match_the_bundle_at_machine_precision`

Quantity	Source	Equation	File	Object	Checked by
relative-sign control: the $c^2 = -1$ normalisation, run as a wiring check	Annals II	(F.1), p. 379	`functions/harmonics/free_streaming.py`	`free_stream`	`test_sign_control_fails_in_a_known_shape`
$\int_0^\infty z^{-m} j_\ell^2(z)\, dz$, corrected to carry $[(m/2)!]^2$	Annals II	(199), p. 372	`scripts/derive_ell_range.py`	`bessel_integral_corrected`	`test_corrected_bessel_identity_matches_quadrature`
$\ell_A = \pi\Delta\eta_*/r_s$, bounding criterion 5's range	Annals II	§8.3.2, p. 372	`scripts/derive_ell_range.py`	`acoustic_scale`	—
$(a'/a)^2 = H_0^2[\Omega_r a^{-2} + \Omega_m a^{-1} + \Omega_k + \Omega_\Lambda a^2]$	this work	the $1+3$ energy constraint	`functions/background/flrw.py`	`Background`	`test_the_gauss_constraint_sign_is_the_corrected_one_and_the_printed_one_has_no_solution`
$\mathcal{P}(k) = \mathcal{A}(k\eta_0)^{n-1}$	Annals II	(J.1)	`functions/spectra/transfer.py`	`power_spectrum_j1`	`test_j1_is_constant_at_n_equals_one_and_eta0_drops_out`
$\mathcal{P}(k) = \mathcal{B}k^n T^2(k)$	Annals II	(J.3)	`functions/spectra/transfer.py`	`power_spectrum_j3`	`test_j1_and_j3_differ_by_one_power_of_k`
$T(k) = [1 + (ak + (bk)^{3/2} + (ck)^2)^\nu]^{-1/\nu}$	Annals II	(J.5)	`functions/spectra/transfer.py`	`transfer_function`	`test_b10_the_printed_transfer_function_diverges_where_it_must_vanish`
$\dot\delta_\ell = -n_e\sigma_T\delta_\ell - ikT\{\frac{\ell}{2\ell-1}\delta_{\ell-1} + \frac{\ell+1}{2\ell+3}[1 - \frac{\ell(\ell+2)K}{k^2}]\delta_{\ell+1}\}$	Wilson	(8), p. 4	`functions/harmonics/external.py`	`integrate_wilson`	`test_wilson_reproduces_the_bundle_and_tests_the_phase_convention`
$x_e(z)$, equilibrium Saha	this work	standard; not in the antecedent	`functions/spectra/decoupling.py`	`saha_ionisation`	`test_saha_is_monotone_and_spans_the_full_range`
$\mathcal{V}(k,\eta) \simeq \kappa' e^{-\kappa}$	Annals II	§5.2	`functions/spectra/decoupling.py`	`RecombinationHistory`	`test_the_visibility_is_a_normalised_probability_density`
$\mathcal{D}(\eta_0,k) = \int d\eta\, \mathcal{V} e^{-(k/k_D)^2} \simeq e^{-(k/k_D)^2}$	Annals II	(101), §5.2	`functions/spectra/decoupling.py`	`RecombinationHistory`	`test_the_visibility_is_a_normalised_probability_density`
the integral solution with $e^{-\kappa}$ on gravitational and $\kappa' e^{-\kappa}$ on scattering sources	Annals II	(98), §5	`functions/spectra/decoupling.py`	`Weighting`	`test_the_two_weightings_are_distinct_objects`
the same solution with both weights replaced by $\mathcal{V}$	Annals II	(102), §5	`functions/spectra/decoupling.py`	`Weighting`	`test_the_visibility_weighting_cannot_carry_a_late_isw`

Quantity	Source	Equation	File	Object	Checked by
$\gamma \simeq \frac{k^2}{6\kappa'} \frac{R^2+\frac{4}{5}(1+R)}{(1+R)^2} = k^2/k_D^2$	Annals II	(168), §6	functions/spectra/decoupling.py	diffusion_scale	test_damping_suppresses_small_scales_and_leaves_large_ones
$-C_0 = \frac{5}{2}a\mathcal{B}_2$, $-C_1 = \frac{1}{k}(a\mathcal{B}_1+\kappa' v_B)$, $-C_2 = \frac{1}{k^2}a\mathcal{B}_2$	Annals II	(90), §5	—	—	—
$\tilde{\tau}_1' = -\frac{R'}{1+R}\tilde{\tau}_1 - \frac{k}{1+R}\delta\tilde{T} - k\Phi_A$; $\delta\tilde{T}' = -aH\Phi_A - \Phi_H' + \frac{k}{3}\tilde{\tau}_1$	Annals II	(152), (153), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$\delta\tilde{T}'' + \frac{R'}{1+R}\delta\tilde{T}' + k^2 c_s^2 \delta\tilde{T} = -\Phi_H'' - \frac{R'}{1+R}\Phi_H' - \frac{k^2}{3}\Phi_A$	Annals II	(154), §7.1	functions/spectra/acoustic.py	solve_acoustic	test_the_pair_reduces_to_the_printed_oscillator_exactly
$c_s^2 = 1/(3(1+R))$, $R = 3\rho_M/4\rho_R$	Annals II	§6	functions/spectra/acoustic.py	sound_speed_squared	test_sound_speed_limits
$\delta\tilde{T} + \Phi_A \approx [\delta\tilde{T}(0) + (1+R)\Phi_A]\cos(kr_s) - R\Phi_A$	Annals II	(180), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_the_integration_converges_to_the_closed_form_180_as_r_is_held_constant
$[\delta\tilde{T} + \Phi_A](\eta_*) \approx \frac{1}{3}\Phi_A(0)(1+3R_*)\cos(kr_s^*) - R_*\Phi_A(0)$	Annals II	(181), §7.1.2	functions/spectra/acoustic.py	sachs_wolfe_at_last_scattering	test_b8_equation_181_sign_is_fixed_by_its_own_limit
$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} = S_P j_\ell(k\Delta\eta_*) + \int d\eta\,(S_{\rm DISW} + S_{\rm ISW}) j_\ell(k\Delta\eta)$	Annals II	(176), §7.1.1	functions/spectra/sources.py	temperature_transfer	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_P = D(\eta_0, k)\,[(\tilde{\delta}_T + \Phi_A) + \kappa' v_B]$	Annals II	(177), §7.1.1	functions/spectra/sources.py	source_primary	test_176_with_an_impulsive_source_is_the_free_streaming_projection
$S_{\rm DISW} = V e^{-(k/k_D)^2}[\frac{1}{3}k\tilde{\tau}_1 + (\kappa'\tilde{v}_B' + \kappa''\tilde{v}_B)]$	Annals II	(178), §7.1.1	functions/spectra/sources.py	source_doppler	test_the_printed_doppler_bracket_equals_the_contracted_form
$S_{\rm ISW} = V e^{-(k/k_D)^2}[(\Phi_A' - \Phi_H') - aH(\tilde{\delta}_T + 3\Phi_A)]$	Annals II	(179), §7.1.1	functions/spectra/sources.py	source_integrated	test_b6_sign_alternative_is_not_a_no_op
$C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{dk}{k} k^3 \lvert\tau_\ell\rvert^2$	Annals II	(186), p. 368	functions/spectra/angular.py	cl_mode_route	test_the_harmonic_normalisation_cancels_out_of_186_exactly
$\langle \tau_{A_\ell} \tau^{A_\ell} \rangle = \frac{\beta_\ell}{2\pi^2}\int k^2 dk \lvert\tau_\ell\rvert^2$	Annals II	(187), p. 368	functions/spectra/angular.py	multipole_mean_square	test_188_is_delta_over_2l_plus_1_times_187

Quantity	Source	Equation	File	Object	Checked by
$C_\ell = \Delta_\ell (2\ell+1)^{-1} \langle \tau_{A_\ell} \tau^{A_\ell} \rangle$	Annals II	(188), p. 368	`functions/spectra/angular.py`	`cl_covariant_route`	`test_the_two_routes_agree_to_machine_precision_on_real_input`
$C_\ell^{\mathrm{SW}} = \frac{1}{2\pi} H_0^4 \Omega_0^{1.54} \int \frac{dk}{k^2} P(k) j_\ell^2(k\Delta\eta_*)$	Annals II	(198), p. 371	`functions/spectra/sachs_wolfe.py`	`cl_sachs_wolfe_quadrature`	`test_the_analytic_reduction_matches_the_quadrature`
large-scale closed form, $n = 1$	Annals II	(200), (201), p. 372	`functions/spectra/sachs_wolfe.py`	`cl_large_scale_reduced`	`test_b5_printed_201_is_short_by_the_comoving_distance`
closed-form k-integral kernel at complex m, $\int z^{-m} j_\ell^2(z) dz$	Annals II	(199), p. 372, corrected	`functions/spectra/kernels.py`	`bessel_square_kernel`	`test_complex_kernel_reduces_to_the_real_identity`
primary-term k-integral by power-law decomposition	Annals II	(176) primary term, (198)	`functions/spectra/kernels.py`	`fftlog_primary_integral`	`test_fftlog_matches_quadrature_on_a_non_power_law_integrand`
integrated terms, $\int d\eta\, S(k,\eta)\, j_\ell(k\Delta\eta)$	Annals II	(176), p. 365	`functions/spectra/bessel_table.py`	`line_of_sight`	`test_line_of_sight_matches_direct_evaluation`
truncation condition: four consecutive vanishing harmonics force $\sigma_{ab} = 0$	Annals II	App. E, (E.1), p. 378	—	—	—

Table 2: Notation, and the names the quantities carry in the code. The mode/multipole distinction is the one that matters most: *Annals II* notes that it is not made in Bardeen-variable treatments.

Symbol	In code	Meaning	Note
Φ_A, Φ_H	`phi_a, phi_h`	the two gauge-invariant potentials entering (177) and (179)	—
$\tilde{\delta}_T$	`delta_T`	covariant temperature perturbation	threading-dependent; $\tilde{\delta}_T + 3\Phi_A$ is the combination in dispute (B-6)
$\tilde{\tau}_1$	`tau_1`	the dipole, which supplies the Doppler term of (178)	—
$\tilde{v}_B$	`v_b`	baryon velocity	—
κ', κ''	`kappa_prime`	differential optical depth and its conformal-time derivative	supplied, never manufactured inside the source terms
V	`visibility`	visibility function, weighting the integrated sources	—
k_D	`damping_k`	diffusion damping scale, entering (178) and (179) as $e^{-(k/k_D)^2}$	—
aH	`conformal_hubble`	a'/a, the background combination (179) carries	derived from the $1+3$ constraint, not from a printed expansion law
S_P, S_{DISW}, S_{ISW}	`source_primary, source_doppler, source_integrated`	the primary, Doppler and integrated temperature sources of (177)–(179)	individually frame-dependent
$\tau(x, e)$	—	temperature anisotropy as a function of position and direction	—
τ_{A_ℓ}	—	**multipole** coefficients — the covariant objects	valid for general geometries; what every released figure plots
τ_ℓ	`tau, values`	**mode** coefficients	almost-Robertson–Walker only. Conflating these two is how a sign is lost
O^{A_ℓ}	—	$e^{\langle A_\ell \rangle}$, the PSTF basis on the sphere	Annals I (15)
Q_{A_ℓ}	—	harmonic basis tensor, $(-k_{\mathrm{phys}})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	the adopted normalisation; see criterion 3
β_ℓ	`beta`	$O^{A_\ell} O_{A_\ell}$	Hu & Sugiyama's normalisation
α_ℓ	`alpha`	$(2\ell+1)/\beta_\ell$	Ma & Bertschinger's and Seljak & Zaldarriaga's normalisation
k_{com}	`k_com`	comoving wavenumber	**never a bare `k` in this codebase**
k_{phys}	`k_phys`	physical wavenumber, k_{com}/a	Appendix F's k is comoving; the conventions sheet's is physical
η	`eta`	conformal time, $dt = a\, d\eta$	—
$\Delta\eta_*$	`d_eta`	$\eta_0 - \eta_*$	—
j_ℓ	`spherical_jn`	spherical Bessel function	the free-streaming solution
u^a	—	the threading 4-velocity	carried generic; frames are specialisations made at the end
σ_{ab}	—	shear	σ is the shear and nothing else
A_a	—	acceleration, $\dot{u}_a$	—
Θ_ℓ	—	Hu & Sugiyama's multipole variable	β-normalised

3 The acceptance spine

Appendix F of the antecedent, pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation

without introducing a representation, and it is where this bundle's independence lies.[3]

Five external treatments are read and **four** are drawn — Hu & Sugiyama Eq. (6), Wilson (1983) Eq. (8), Ma & Bertschinger Eqs. (49) and (50), and Seljak & Zaldarriaga Eq. (3d) — each transcribed from its own paper.[4] The fifth is Wilson & Silk (1981) Eq. (7), the earliest of them and likewise read from its own page: at $K = 0$ it *is* Wilson (1983) Eq. (8), so it is credited and not drawn, four panels carrying four independent numerical checks. Hu & Sugiyama and Wilson write the recursion in the β normalisation — the normalisation their Eq. (2) fixes — and the other two in the α normalisation, which is why Appendix F prints both (F.3) and (F.4). A basis phase leaking into the couplings would break at least one of the two matches.

Wilson is the strongest of the four, and not because of his weights, which are Hu & Sugiyama's. He writes the recursion in the *imaginary* convention, with both bracket terms positive under an overall $-ikT$, where the others carry opposite signs under a real k. The two are related by $\Theta_\ell = i^\ell \delta_\ell$, which turns one form into the other exactly. He is integrated complex and as printed, and the phase removed only afterwards; transforming the equation first would assume the identity under test. That the transformed solution is *real* to 10^{-10} is a separate assertion from that it agrees. Wilson therefore tests the phase convention — the subject of §2's footnote and of acceptance criterion 3 — from a source published before any of the others.[5]

4 Conventions, and what the comparison does and does not settle

The same hierarchy is written four different ways in the literature this bundle checks against, and the differences are not cosmetic. Table 3 is the translation, generated from `config/implementation-register.toml` so that it cannot drift from the code, with the test that checks each row.

There are exactly two conventions, and they differ by $c^2 = -1$. Changing the harmonic normalisation $Q_{A_\ell} = \mu^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$ rescales the mode coefficients as $\tau_\ell \to c^\ell \tau_\ell$ with $c = \mu'/\mu$. A recursion $\tau'_\ell = k[A_\ell \tau_{\ell-1} + B_\ell \tau_{\ell+1}]$ then carries $(A_\ell/c,\, B_\ell c)$, so the ratio B/A picks up c^2. At $c = i$ an opposite-sign real bracket becomes a same-sign bracket under an overall i. Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga print the first; Wilson prints the second. That is the whole of the difference, and it is asserted symbolically rather than described.

What the match establishes. The ℓ-weights, which are convention-independent, are confirmed by five sources in two normalisations, four of them drawn and numerically independent. The bundle is shown to sit at the same c^2 as Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga, and Wilson is shown to sit at the other. *Both* Wilson papers are in the imaginary convention, which settles a standing hypothesis in the negative: no author published this recursion in both conventions, so the split is between authors and not within one.

[3] The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded.

[4] Ma & Bertschinger print the same recursion twice. Eqs. (49) and (50) are the *massless neutrino* hierarchy, synchronous and conformal Newtonian, and are collisionless; their Eqs. (63) and (64) are the photon hierarchy, carrying the identical recursion plus a Thomson term $-an_e\sigma_{\mathrm{T}}F_{\gamma\ell}$. Appendix F's (F.4) is a free-streaming recursion, so (49) and (50) are the comparison and (63) and (64) are not. Published and arXiv numbering agree for this paper, which is not the case for the Annals papers.

[5] Appendix F's citation pairs the name of one paper with the key of another. Both papers are real and both numbers are right for the paper they belong to: Wilson & Silk (1981) Eq. (7) *is* the $\ell > 2$ hierarchy, and Wilson (1983) Eq. (7) is that paper's quadrupole equation while its Eq. (8) is the hierarchy. The prose names Wilson & Silk, the key is Wilson alone and consistent across every citation, and the load-bearing $i^{-\ell}$ remark names Wilson alone — so the match and the phase remark are about Wilson (1983), and the number beside them is correct for the 1981 paper the prose names. One mismatched pairing, not two independent misprints; recorded, and it moves no result.

One source fixes its own phase, and that matters. Ma & Bertschinger define their Legendre expansion with an explicit factor $(-i)^\ell(2\ell+1)$ and state why: it is "chosen to simplify the expansion of a plane wave", so that $\exp(-i\mathbf{k}\cdot\mathbf{x})$ has coefficients $F_\ell = j_\ell(kr)$. Their coefficients therefore sit at $(-i)^\ell$ from the plane wave *by their own definition*, are real, and satisfy the opposite-sign real recursion. For this source the phase is read off the paper rather than assumed, which is exactly the standing the next paragraph says three of the four lack.

What it does not establish, and this release says so. It does **not** establish that $Q_{A_\ell} = (-k_{\rm phys})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ is the right reading of *Annals I*'s definition. Rephasing *both sides* of a comparison by i^ℓ is a change of variable and cannot fail — measured at 6.2×10^{-15} either way — so a match constrains the convention only where the phase applied to the external source is fixed from outside the comparison. For two of the four, no phase is applied, and that is a choice rather than a derivation; Ma & Bertschinger and Wilson are the two that are fixed from outside the comparison.

The exception, and it is the useful one. *Annals II* states, in the line following its flat-case mode functions, that the covariant form "differs by a factor of $i^{-\ell}$ from Wilson since we are using plain mode functions instead of plane waves". That phase is specified by the antecedent, not chosen here. So Wilson's panel in Figure 1 is the one comparison carrying information about the convention, and it is checked in two independent ways: the transformed solution agrees with the reconstruction, *and* it is real to 10^{-10}, which the agreement alone would not show.

Acceptance criterion 3 was therefore **reopened** and restated to the claim the evidence supports. Settling it needs *Annals I*'s definition of Q_{A_ℓ}, not a better match. Nothing computed here depends on it: the covariant τ_{A_ℓ} is invariant under the choice, the mode coefficients are not, and every released figure plots the invariant quantity.

Table 3: How each treatment in the foundational literature writes the same hierarchy, and how this bundle checked it. The two conventions in the literature are related by $\tau_\ell \to c^\ell \tau_\ell$ with $c^2 = -1$ and nothing else. **Where the transformation applied is "none" that is a choice, not a derivation** — which is why acceptance criterion 3 was reopened; see §4.

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Annals II (F.1), Appendix F	τ_ℓ	$\frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}$ on $\tau_{\ell+1}$, 1 on $\tau_{\ell-1}$	covariant; $Q_{A_\ell} = (-k_{\text{phys}})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$	—, this is the reference	arithmetic on β_ℓ alone; no representation introduced
Annals II (F.3), Appendix F	$\beta_\ell \tau_\ell$	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none	(F.1)→(F.3) by multiplying through by β_ℓ, as exact rationals
Annals II (F.4), Appendix F	$\alpha_\ell^{-1} \tau_\ell$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	the $(2\ell+1)^{-1}$ the prose states and the display omits	**finding B-2**: printed (F.4) is not yet in the Ma & Bertschinger form
Hu & Sugiyama Eq. (6), p. 2601	Θ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper, never through Appendix F
Wilson Eq. (8), p. 4	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**; both bracket terms + under an overall $-ikT$	$\Theta_\ell = i^\ell \delta_\ell$ — **fixed by the antecedent**, which states the $i^{-\ell}$ itself	integrated *complex and as printed*; the phase removed only afterwards, and the result asserted real to 10^{-10} separately from that it agrees
Wilson & Silk Eq. (7), p. 15	δ_ℓ	$\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$	**imaginary**, as Wilson 1983 — *not* the real form	none required: at $K = 0$ it is Wilson 1983 (8) exactly	read from its own page. The **earliest** of the five and a fifth source for the weights, but **not** a fifth numerical check — the same equation cannot check itself
Ma & Bertschinger Eqs. (49), (50)	$F_{\nu\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed from its own paper; synchronous and conformal Newtonian gauges

Treatment	**Variable**	**Bracket weights**	**Phase convention**	**Transformation applied here**	**How it was checked**
Seljak & Zaldarriaga Eq. (3d)	$\Delta_{T\ell}$	$\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$	real, opposite-sign bracket	none — *a choice*, see Q1	integrated as printed; also the method CMBFAST implements, cited for criterion 5b

The second table, and what its two thesis columns are for. Table 3 translates between the treatments this bundle is *checked against*. Table 4 does something different: it writes each coupling of this reconstruction in the conventions of Maartens, Gebbie & Ellis (1999), Challinor (2000), the thesis in **both** of the conventions it uses, Beneke & Fidler (2010) and Pitrou (2009). Every cell is derived from a named line of the normative sheet, and no cell is blank: where a treatment has no established counterpart the cell says so.

Two things in it are worth reading rather than skimming. **The thesis takes two columns and they differ on exactly the kinematic rows** — its $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping the overall sign against its K/Π form. Elsewhere the two agree and the table says so in as many words, because a difference in wording where there is none in substance is precisely how two conventions get conflated while appearing distinct. And **the Beneke & Fidler and Pitrou columns are mostly "not established"**: the sheet records both as unsettled at the level of hierarchy coefficients, Beneke & Fidler working in (ℓ, m) tetrad form and Pitrou in Poisson gauge, and neither has had a hierarchy coefficient reproduced here. That is the honest content of those columns and it is printed rather than left to inference.

Table 4: The couplings of this reconstruction written in each external convention. Every cell is derived from the line of `provenance/conventions.md` named in the *Sheet* column; no cell is blank, and a treatment with no established counterpart says so rather than being left to inference. **The thesis takes two columns and they differ on exactly the kinematic rows** — the $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping its overall sign against the K/Π form. On the remaining rows the two agree, and the table says *as Th-K* rather than restating them: a difference in wording where there is none in substance is how two conventions get conflated while appearing distinct.

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Rank raising	`C4.2`	identity	identity in form; D_a unchanged, the sign sits in k	identity	as Th-K	not established: BF work in (ℓ, m) tetrad form, no coefficient reproduced	not established: Pitrou works in Poisson gauge, no hierarchy coefficient checked
Rank lowering	`C4.4`	identity	identity in form; CL (2.25) prints $+\mathrm{D}^b I_{bA_\ell}$, reproduced from MGE99 (71)	identity; Th App. D footnote to (D.44) at $\ell = 1$ agrees	as Th-K	not established	not established
Free-streaming mode recursion	`C4.5`	identity	identity in form; CL (2.25) free streaming reproduced from MGE99 (71)	identity	as Th-K	not established	not established
PSTF contraction on the sphere	`C6.1`	identity (one source: EMT 1983, via C6.4)	identity; CL expands in the same O^{A_ℓ}	identity, Th (1.20)	as Th-K	not established: BF use ${}_sY_{\ell m}$, not a PSTF basis	identity at the basis level: Pitrou (1.28) expands in n^{A_ℓ}
PSTF self-contraction β_ℓ	`C5.2`	identity	identity; CL absorbs it into Δ^C_ℓ, see the row below	identity, Th (1.23)	as Th-K	no counterpart: (ℓ, m) normalisation carries $\sqrt{4\pi/(2\ell+1)}$ instead	identity, Pitrou (1.29) in the equivalent $4\pi\ell!/(2\ell+1)!!$ form
Angular measure Δ_ℓ	`C5.5`	identity, MGE99 (47), same value	$\Delta^C_\ell = 4\pi(-2)^\ell(\ell!)^2/(2\ell+1)! = (-1)^\ell\Delta_\ell$, CL (2.23)	identity	as Th-K	no counterpart in this form	identity, Pitrou (1.29), verified symbolically

Coupling	**Sheet**	**MGE99**	**CL**	**Th, K/Π**	**Th, $\dot{\mathcal{T}}$**	**BF**	**P09**
Shear coupling at $\ell+2$	`C9, T-MGE and T-Th-T`	$-(\ell+2)$ as printed in (71), (88), (90), (99) — **defect E9**; $+(\ell+2)$ is correct, and MGE99 (89) at $\ell=2$ prints $+\frac{8}{15}\rho\sigma$, agreeing with the correction	$-\frac{8}{15}\delta_\ell^2 I\sigma$ gives $+(\ell+2)$, independently of MGE99	$-(\ell+2)$ in Th (2.73) — carries E9	$+(\ell+2)$ **on the RHS** in Th (2.84): every kinematic term flips overall sign against (2.73) because it is moved across. **This row is where the two thesis columns differ**	not established: BF put the observer at rest, so they carry no shear coupling	not established
Acceleration coupling at $\ell\pm1$	`C9, T-CL and T-Th-T`	identity	$A_a \to -A_a$ under $g \to -g$ at fixed u^a (CL footnote to §2), the signature difference and nothing more	identity	overall sign flipped with the rest of the kinematic terms — **differs from Th-K**	not established: observer at rest	not established

5 Frames and specialisation

The solution is carried with a **generic** u^a. The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation *Annals II* itself makes at p. 365 in order to remove the shear terms, and the **total-energy frame** $q_a = 0$. Carrying the generic threading and specialising late is the difference the $1+3$ covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

"CDM" names two different things, and this supplement keeps them apart. *CDM the model* is a cosmology — flat, matter dominated, no Λ — and it is one of the two models this release delivers a spectrum for. *The CDM frame* is a choice of threading u^a, and it is the total-energy frame $q_a = 0$: cold dark matter is pressureless and geodesic and carries no energy flux of its own, so the threading in which the CDM is at rest is the one in which the total energy flux vanishes. The two are independent. The ΛCDM *model* is computed in the same *frame*, and a change of model is not a change of frame. In the code the models are named `CDM_MODEL` and `LCDM_MODEL` and the frames `"energy"` and `"newtonian"`, so the two cannot be confused at a call site.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, **v1.5.0's restricted case is a recovery and a check** — it must return v1.0.0's numbers, and that requirement is the regression test rather than a new result.

The covariance claim is established symbolically, and that is the stronger form. Acceptance criterion 2 asks that a change of threading $u^a \to \tilde{u}^a = u^a + v^a$ return the rank-one bracket unchanged up to $\dot{v}_a$. It does, *identically* in exact arithmetic rather than to a tolerance, and the cancellation is exhibited rather than quoted:

- $\tilde{\mathrm{D}}_{\dot{a}} \ln T = \mathrm{D}_a \ln T - H v_a$, the extra term arising because the projector moves with the frame and $\overline{\ln T} = -H$ in the background;
- $\tilde{A}_a = A_a + \dot{v}_a + v^b \nabla_b u_a$, where $v^b \nabla_b u_a$ is formed *in full* — $H v_a$ together with $\sigma_{ab} v^b$ and $\epsilon_{abc} v^b \omega^c$ — and the latter two leave at $O(\varepsilon^2)$ by series expansion rather than by hand;
- so $\tilde{\mathcal{B}}_1 = \mathcal{B}_1 + \dot{v}_a$ with **no shear and no** Hv. It is the *same* H twice, which is why the cancellation is structural and not a coincidence of two fitted coefficients.

The curvature half. The deflection depends only on $\Phi_A - \Phi_H$, whose trace-free screen-projected Hessian is the electric Weyl tensor, $E_{ab} \simeq \frac{1}{2}\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}(\Phi_A - \Phi_H)$. Since E_{ab} vanishes in the background, the Stewart–Walker lemma makes it frame-invariant at first order, and therefore so is $\Phi_A - \Phi_H$. Both potentials shift by the same $(v' + \mathcal{H}v)/k_{\mathrm{com}}$, and the residual $\tilde{E}_{ab} - E_{ab}$ is identically zero — **with a negative control**: a rule leaving Φ_H fixed, or flipping its sign, leaves a residual proportional to $(v' + \mathcal{H}v)$ and fails. The invariance is therefore forced rather than chosen to make a residual vanish. Only the dipole moves, $\tilde{\tau}_a \simeq \tau_a - v_a$; every τ_{A_ℓ} with $\ell \geq 2$ is unchanged, which is the standard any threading dependence surviving at $\ell \geq 2$ must meet: it is an artefact of the reduction, not physics.

The energy frame is derived, not stipulated. Imposing $\tilde{A}_a = 0$ gives $v' + \mathcal{H}v = -k_{\mathrm{com}}\Phi_A$ — the Euler equation of a pressureless geodesic fluid, which is what the cold dark matter obeys. The normalisation is corroborated independently: in the baryon-dominated limit $R \to \infty$, Eq. (152) reduces to that same equation, so the photon dipole and the frame velocity carry one normalisation.

What this release does *not* do, stated because the earlier draft of this section claimed it. It does not compute an assembled spectrum in two frames and draw the difference. Eqs. (177)–(179) are **Newtonian-threading expressions**: they descend from (106), which the corrections record flags in those words, and the Newtonian threading carries its whole $O(\ell)$ source in the acceleration pair while the CDM threading carries its whole source in the shear pair. Transforming the four input fields and re-evaluating the same equations therefore deletes the coupling that was carrying the source without supplying the one that replaces it: measured, the transfer function moves by a factor of about five, and under the opacity weighting the source collapses to two parts in a thousand. That is not a frame dependence of the sky. The figures which would draw it, and the convergence figure which would test footnote 29 on p. 365 in both threadings, require the generic-u^a hierarchy and are deferred with it.

6 Comparison and its range

The C_ℓ comparison is Sachs–Wolfe only: (196) descends from (181), in which the acoustic modulation $\cos(kr_s)$ is dropped by taking $r_s^* \to 0$, and §8.3.2 sets the transfer function to unity. Its accuracy is therefore bounded by the term that was dropped. The range is derived, not chosen: $2 \le \ell \le 20$ at 5%, generated by `scripts/derive_ell_range.py`.

The wider comparison cites CMBFAST for its *method* — line-of-sight integration, Seljak & Zaldarriaga, the same reference the antecedent's own bibliography carries — and compares against **CAMB** and **CLASS**, because the Fortran 77 original is unmaintained and effectively unrunnable. Saying so is more honest than claiming a comparison against a code nobody can run. The tolerance is *measured* rather than chosen: the CAMB–CLASS mutual difference is the floor, since no agreement can be claimed tighter than two trusted independent codes agree with each other. It is reported in three bands, whose failure modes differ — $2 \le \ell < 30$, cosmic-variance dominated and meaningful only binned; $30 \le \ell < 1000$, the acoustic peaks, where a hierarchy or recursion error actually shows; and $1000 \le \ell \le 2000$, the damping tail, where C_ℓ is small and fractional differences blow up on a vanishing signal. The comparison is made against the corrected closed form.[6][7]

The external codes of criterion 5b are compared in `provenance/EXTERNAL-CODE-COMPARISON-v1.md`, which records what CAMB and CLASS each model and — more to the point — what the comparison structurally cannot settle. Neither carries the $1 + 3$ covariant formalism, neither distinguishes mode from multipole coefficients, and neither carries the $O(\varepsilon^2\ell)$ couplings of v1.5.0. So 5b is a check on the *linear* reconstruction and is incapable of validating the nonlinear extension, which is precisely why v1.0.0 must carry no $O(\ell)$ coupling: if one leaked in, the disagreement with CAMB and CLASS would be indistinguishable from a bug. The canonical parameter set switches reionisation, lensing and massive neutrinos **off**, each being physics this bundle does not model.

7 Evidence

Each figure is generated by the script named in its caption and is accompanied by a standalone caption file in `captions/`. This section is itself generated by `scripts/make_figure_pages.py`, so a figure appears here only once it exists in the tree, and what is outstanding is stated rather than omitted.
Deferred, by recorded decision: **F2** truncation convergence, both frames — v1.2.0, S23; **F5** source decomposition, two frames — v1.2.0, S23; **F7** frame specialisation — v1.2.0, S23. Recorded choices, not

[6] The reduction of (200) carries a factor of the comoving distance that the displayed (201) does not; substituting $z = k\chi$ gives χ^{2-n}, which is χ^1 at $n = 1$. The companion result (204) descends from a $P(k)$ definition differing by one power of k and is correctly free of that factor, so the asymmetry is not a units convention. A misprint, corrected silently and recorded.

[7] That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2 + 1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the D_ℓ of (205) are. A misprint, corrected silently and recorded.

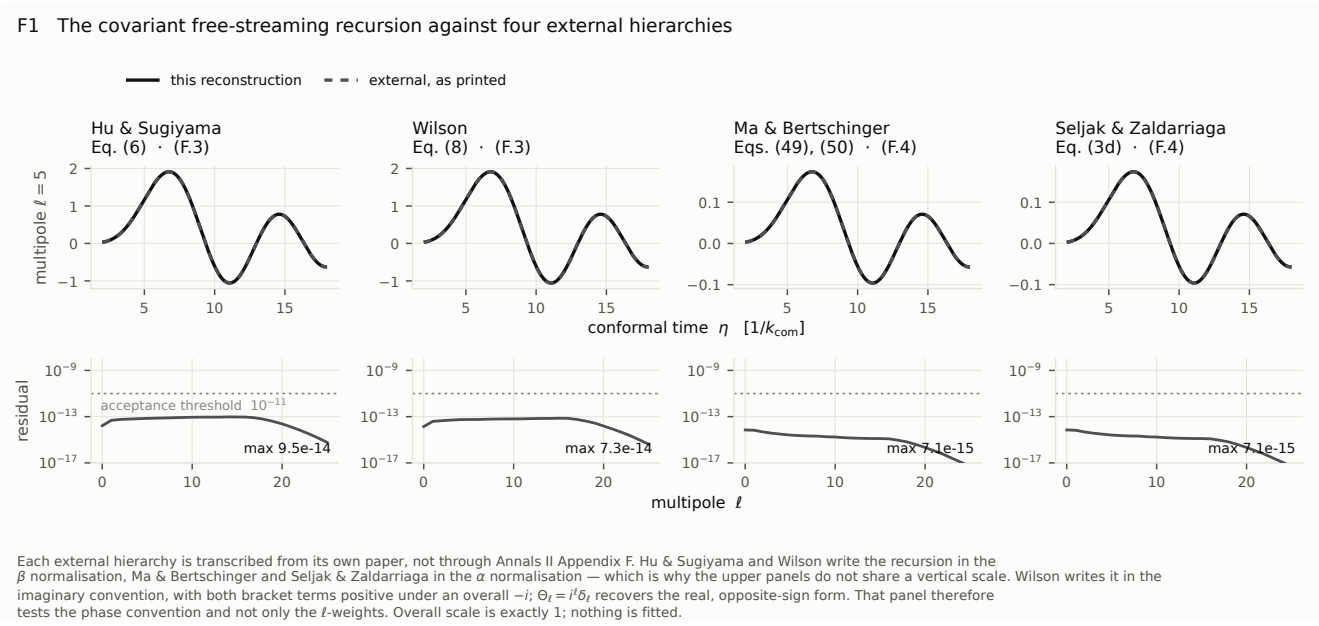


Figure 1: **F1 — The covariant free-streaming recursion against four external hierarchies.** The covariant free-streaming recursion of *Annals II* Appendix F, integrated from a unit monopole, against four external treatments of the same hierarchy. Each is transcribed from its own paper, never through Appendix F. **Upper panels, context:** the reconstruction and the external form at $\ell = 5$; they coincide. The panels do *not* share a vertical scale, because two normalisations are on display. **Lower panels, the claim:** $\max_\eta |\text{external} - \text{reconstruction}|$ against ℓ, with the acceptance threshold marked. Hu & Sugiyama write the recursion in the β normalisation and Ma & Bertschinger and Seljak & Zaldarriaga in the α normalisation, which is why Appendix F prints both (F.3) and (F.4); a basis phase leaking into the couplings would break at least one match. Overall scale is exactly 1.000000000 — nothing is fitted. Four panels from five sources: Wilson & Silk (1981) is the earliest, but at $K = 0$ it is Wilson (1983) Eq. (8) exactly, so it is credited rather than drawn.

Evidence category: cross-check. *Generated by* `scripts/figure_f1_appendix_f.py`. *Standalone caption:* `captions/F1-appendix-f.md`.

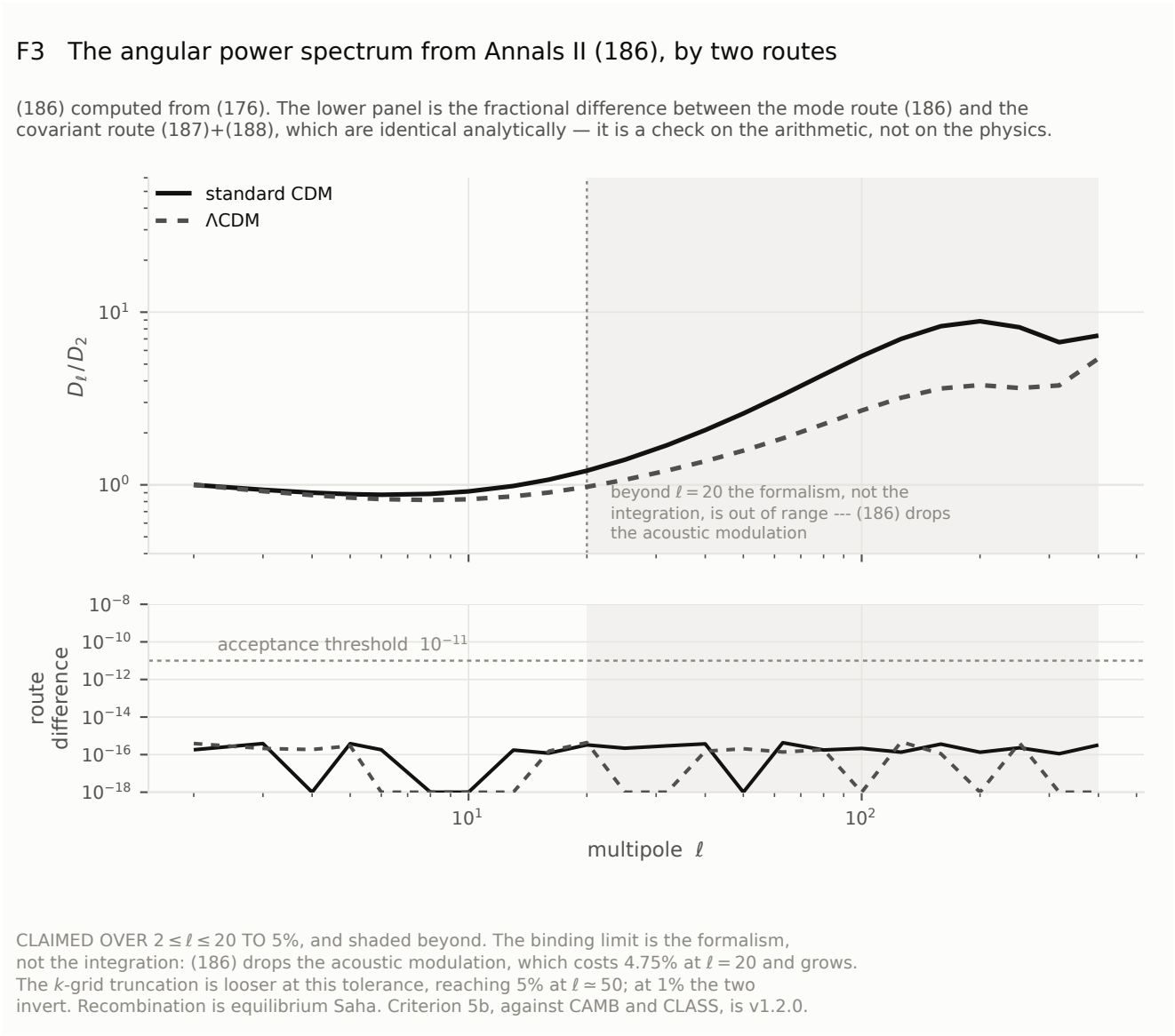


Figure 2: **F3 — The angular power spectrum from Annals II (186), by two routes.** Eq. (186) computed from Eq. (176), for the standard-CDM model of §8.3.2 and for ΛCDM, which *Annals II* does not treat. **Upper panel:** $D_\ell = \ell(\ell+1)C_\ell/2\pi$, each normalised to its own $\ell = 2$ value, so that the comparison is of *shape* and not of an amplitude neither model fixes here. **Lower panel:** the fractional difference between the mode route of (186) and the covariant route of (187) with (188). Those two are identical analytically, so the panel is a check on the arithmetic and not on the physics — and it is worth having, because it caught a factor of four in this bundle's own harmonic weights before any figure was drawn. It sits at 10^{-16}, five orders below the 10^{-11} acceptance threshold.

Evidence category: recovery. *Generated by* `scripts/figure_f3_angular_spectrum.py`. *Standalone caption:* `captions/F3-angular-spectrum.md`.

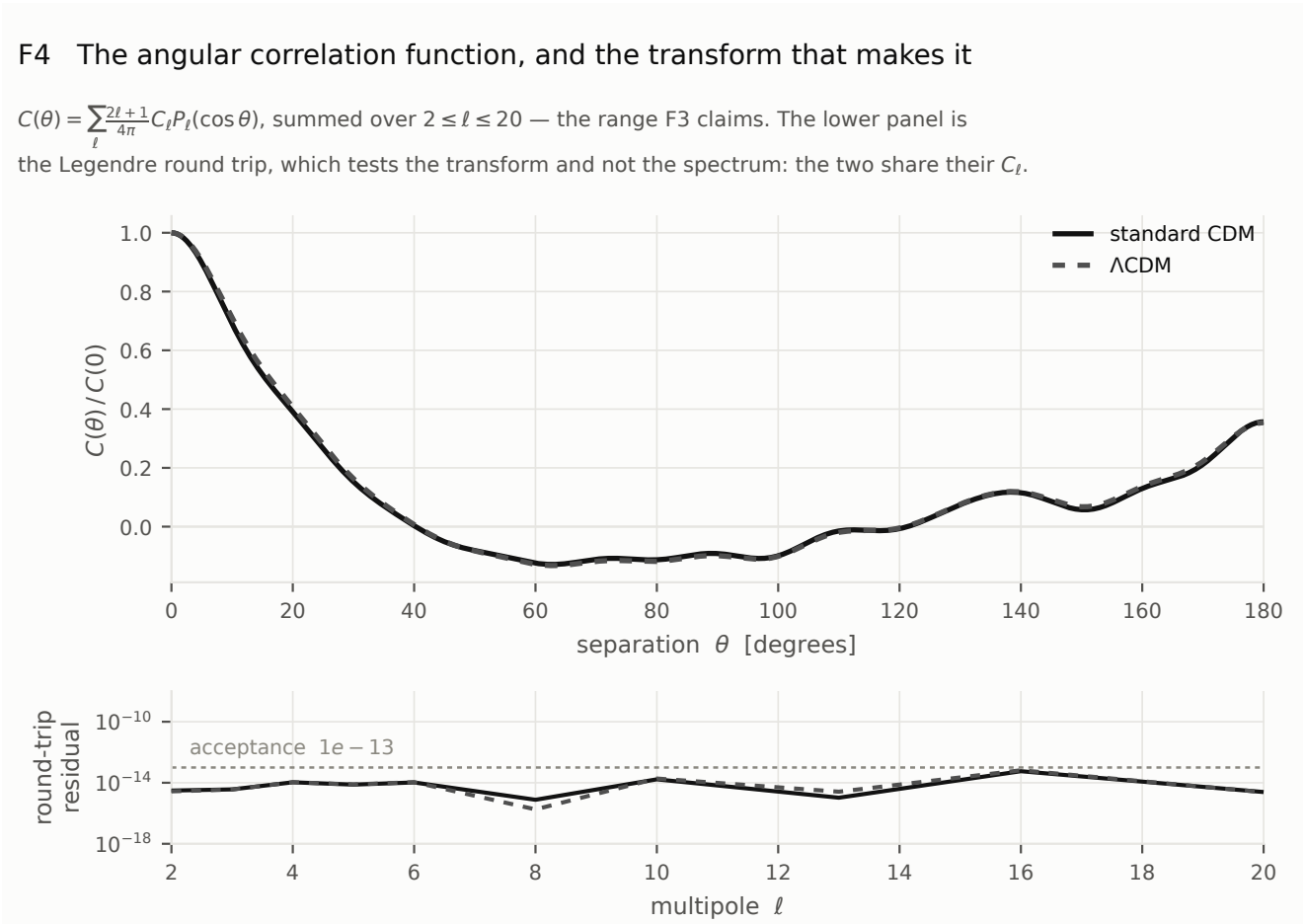


Figure 3: **F4 — The angular correlation function, and the transform that makes it.** $C(\theta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta)$, the real-space view of Figure 3: the same content read as a correlation between two directions separated by θ rather than as power at a multipole. **Upper panel:** $C(\theta)/C(0)$ for standard CDM and ΛCDM, summed over $2 \le \ell \le 20$ — the range Figure 3 claims, bounded by (186) dropping the acoustic modulation. Summing further would draw multipoles this release does not stand behind, and a real-space curve hides which ℓ contributed, so the restriction is made in the transform rather than in this caption. **Lower panel:** the Legendre round trip $C_\ell \to C(\theta) \to C_\ell$, at 4×10^{-14} and 7×10^{-14} against an acceptance of 10^{-13}. It tests the transform and *not* the spectrum — the two share their C_ℓ — which is the same caution Figure 3's residual panel carries and for the same reason.

Evidence category: acceptance. *Generated by* `scripts/figure_f4_correlation.py`. *Standalone caption:* `captions/F4-correlation.md`.

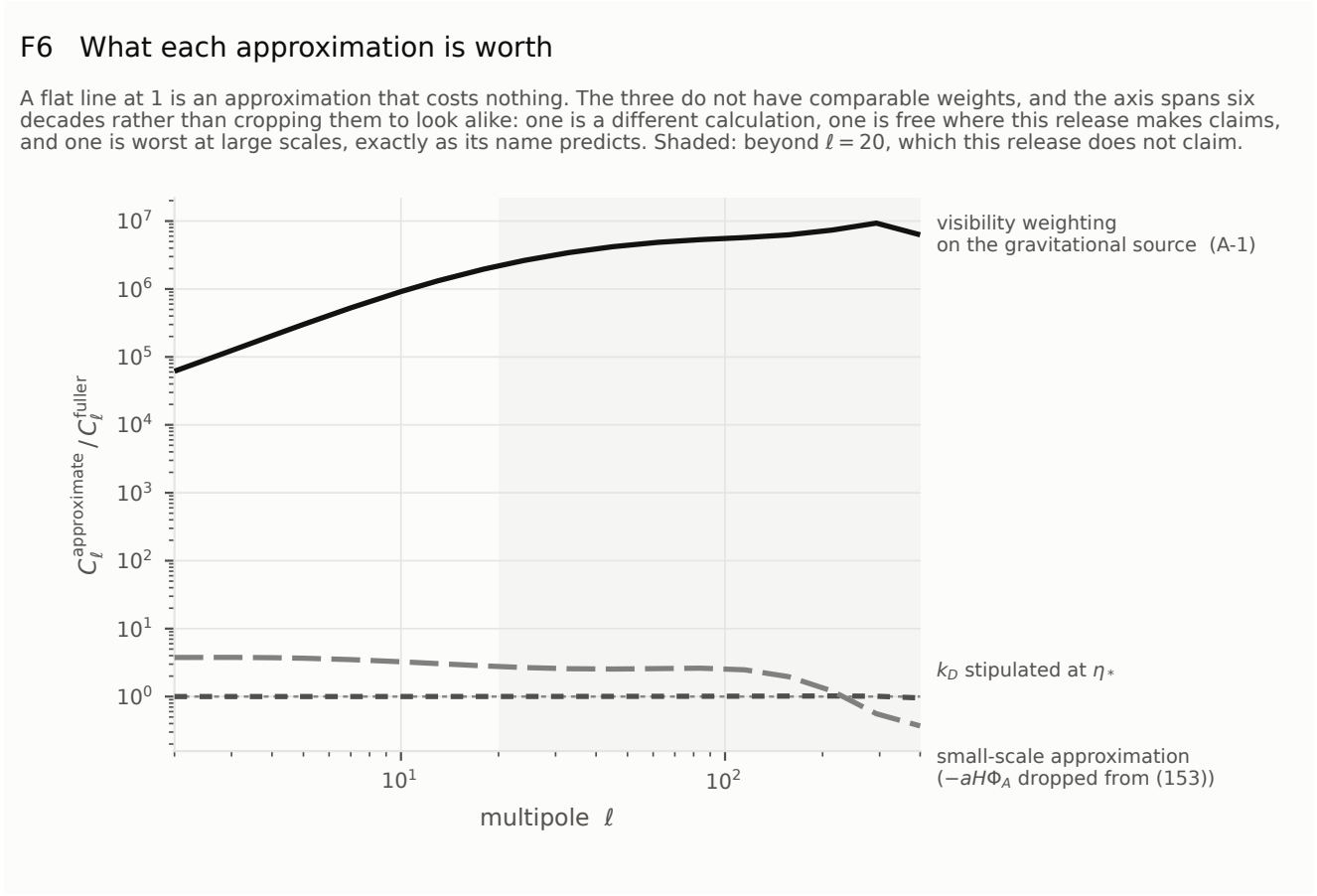


Figure 4: **F6 — What each approximation is worth.** Each series is $C_\ell^{\text{approximate}}/C_\ell^{\text{fuller}}$ for standard CDM, so the panel is scale-free and a flat line at unity is an approximation that costs nothing. *Annals II* makes three narrowing choices which this bundle reproduces rather than corrects, because reproducing a paper means reproducing its approximations (decision S15); they are recorded as **A-n** in `provenance/SOURCE-APPROXIMATIONS-v1.md`, deliberately apart from the defects recorded as **B-n**. This is the one place their cost is a number rather than an adjective.

The three do not have comparable weights, and that is the result, so the axis spans six decades rather than cropping them to look alike. The visibility weighting on the gravitational source (A-1) runs to 10^4–10^6: it is a *different calculation*, not a perturbation of one, which is why `Weighting` has no default. Stipulating k_D at last scattering costs under 0.1% across the claimed range and about 5% by $\ell = 400$ — it **vindicates the release**, which takes the cheap choice where it makes claims. The small-scale approximation, dropping $-aH\Phi_A$ from (153) and so producing the printed oscillator (154), departs from unity most at *large* scales, which is what its name predicts.

A correction kept because it nearly became this figure. The k_D series was first computed with $k_D(\eta)$ along the whole ray. After last scattering $\kappa' \to 0$, so $1/\kappa'$ in (dif-damp) diverges and k_D collapses, damping every integrated source to nothing — a plasma formula evaluated in a universe that no longer has a plasma. That version reported the release's choice as costing a factor of nine at $\ell = 2$; the factor was the artefact. Clamped to $k_D(\min(\eta, \eta_*))$ the comparison is a real one. Shaded: beyond $\ell = 20$, which this release does not claim.

Evidence category: diagnostic. *Generated by* `scripts/figure_f6_approximations.py`. *Standalone caption:* `captions/F6-approximations.md`.

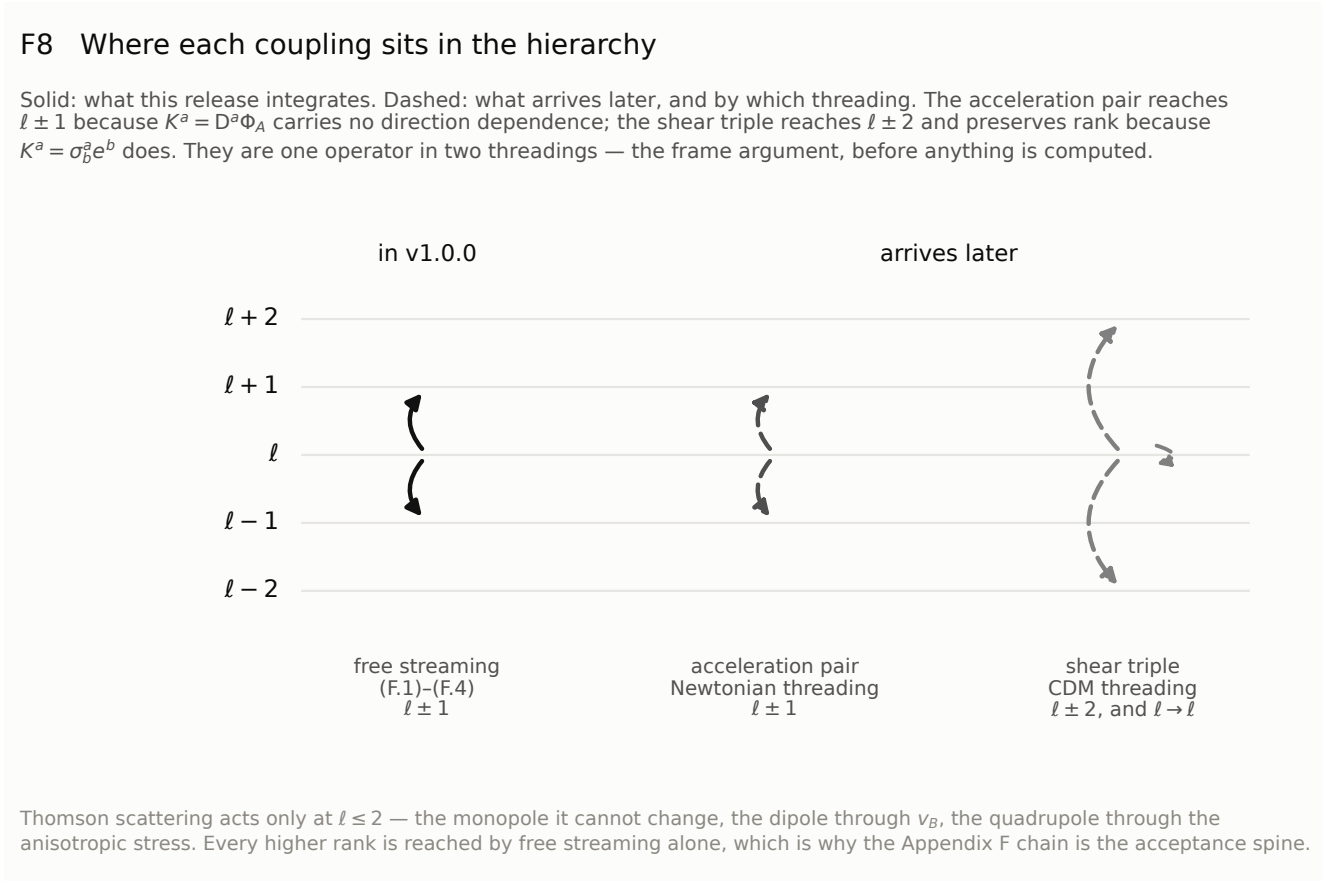


Figure 5: **F8 — Where each coupling sits in the hierarchy.** The one released figure that computes nothing: a statement about the *structure* of the exact divergence hierarchy, read off the equations rather than measured. It is generated by a script like every other figure, so that the manifest covers it — a hand-drawn diagram is the one artefact that can drift without any gate noticing.

Solid: what this release integrates. Free streaming, (F.1)–(F.4), couples rank ℓ to $\ell \pm 1$ and to nothing else, which is the whole of v1.0.0's hierarchy and the reason the Appendix F chain is the acceptance spine. **Dashed:** what arrives later, and by which threading. The acceleration pair reaches $\ell \pm 1$ because $K^a = \mathrm{D}^a\Phi_A$ carries no direction dependence of its own; the shear triple reaches $\ell \pm 2$ *and* preserves rank because $K^a = \sigma^a{}_b e^b$ does. The rank-preserving coupling exists only when K^a depends on direction, and is the single structural difference between the two threadings.

Both are the same operator $K^a_\perp \partial/\partial e^a$, and the number of ranks each reaches is fixed by that one property. **That is the whole content of the frame argument, visible before any of it is computed.** Decision S9 is drawn as well as asserted: every coupling in the solid half is a ratio of linear polynomials and tends to a constant, so an $O(\ell)$ coupling wired in early would fail `tests/test_appendix_f_chain.py` at large ℓ.

Evidence category: schematic. *Generated by* `scripts/figure_f8_schematic.py`. *Standalone caption:* `captions/F8-schematic.md`.

omissions: the reasoning for each is in `provenance/DECISIONS-v1.0.0.md`. The frame-covariance claim F5 would have drawn is established symbolically in this release, as an exact identity, by `functions/frames.py`. The full specification is `captions/FIGURE-PLAN-v1.0.0.md`.

8 Parameter and numerical evidence

Table 5 lists every number the numerics use. Its *provenance* column distinguishes values taken from the source paper, from standard values that are not, from values derived here, and from values stipulated for this release. The distinction matters: $\Omega_\gamma h^2$ is a standard value and not one the antecedent supplies, and it enters only through $R(\eta_*)$ and hence the sound speed.

Table 5: Every number the numerics use. *Provenance* distinguishes values taken from the source paper, standard values that are not, values derived here, and values stipulated for this release.

Symbol	Code	Value	Provenance	Set in
k_{com}	`K_COM`	1.0	stipulated	`scripts/figure_f1_appendix_f.py`
$\ell_{\max}$	`ELL_MAX`	80	stipulated	`scripts/figure_f1_appendix_f.py`
ℓ_{compared}	`ELL_COMPARED`	25	stipulated	`scripts/figure_f1_appendix_f.py`
rtol, atol	`rtol, atol`	10^{-10}, 10^{-12}	stipulated	`functions/harmonics/free_streaming.py`
z_*	`Z_STAR`	1100	standard	`scripts/derive_ell_range.py`
h	`H`	0.5	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
Ω_B	`OMEGA_B`	0.05	AnnalsII §8.3.2	`scripts/derive_ell_range.py`
$\Omega_\gamma h^2$	`OMEGA_GAMMA_H2`	2.47×10^{-5}	standard	`scripts/derive_ell_range.py`
tolerance	`ADOPTED_TOLERANCE`	5%	derived	`scripts/derive_ell_range.py`
acceptance threshold	`--`	10^{-11}	stipulated	`scripts/figure_f1_appendix_f.py`

9 Numerical resolution and reproducibility

The hierarchy is integrated with an explicit Runge–Kutta method of order 8 (`DOP853`) at relative tolerance 10^{-10} and absolute tolerance 10^{-12}, truncated at $\ell_{\max} = 80$ with comparisons restricted to $\ell \leq 25$ so that reflection from the truncation does not contaminate them. The acceptance threshold used in the figures, 10^{-11}, sits an order above the integrator floor.

The release route is

```
python scripts/run_all.py --strict
```

which regenerates every output, figure and table, recomputes the SHA-256 manifests, and fails if any artefact differs from the recorded hash. `FILE-MANIFEST-SHA256.txt` is the release fingerprint; `IMMUTABLE-MANIFEST-SHA256.txt` covers frozen reference material, where any change is release-blocking. The corrections record is append-guarded rather than frozen, because it must take new findings: appending is permitted and rewriting recorded bytes is not.

Weights are carried as exact rationals rather than floating-point numbers wherever a sign or an ℓ-weight is asserted.

10 Beyond this release: the almost-EGS bounds

The Ehlers–Geren–Sachs theorem states that exact CMB isotropy for all fundamental observers in an expanding dust universe forces exact Friedmann–Lemaître–Robertson–Walker geometry. The *almost*-EGS theorem of Stoeger, Maartens & Ellis, *Ap. J.* **443**, 1 (1995) relaxes that to bounds: limits on the CMB multipoles τ_{A_ℓ} become limits on σ_{ab}, ω_{ab}, $\dot{u}_a$, E_{ab} and H_{ab}.

Those bounds are stated in the *covariant multipoles* — not in mode coefficients — which is what this bundle computes and what a gauge-fixed code cannot supply without a translation. Applying them is therefore natural work for this formalism, and it is **v2.0.0** work, because it is a calibration against data rather than an analytical reconstruction.

Table 6: The structure v2.0.0 fills. **No cell is computed at v1.0.0**, and none is quoted here from memory: observational columns are filled from the mission papers at v2.0.0 under the same rule that equation numbers are resolved from the source.

Mission	ℓ **reach**	**bound, standard CDM**	**bound, ΛCDM**
COBE-DMR (1992)	$\lesssim 20$	*the original*	*not in the original*
WMAP (2003–2013)	$\lesssim 1000$	pending	pending
Planck (2013–2018)	$\lesssim 2500$	pending	pending

Why the table needs both columns. The original bounds were derived for a CDM model, not for ΛCDM, and the difference is not cosmetic. The argument converts multipole limits into limits on σ/H using the background expansion, and Λ changes $H(z)$ at low redshift. More sharply, Λ adds a source the original did not have: the late integrated Sachs–Wolfe effect contributes to exactly the low-ℓ multipoles the COBE bounds rest on. In standard CDM it is absent — Φ is constant, which this bundle's growth factor reproduces to six figures — and in ΛCDM it is not, Φ falling to 0.779 of its early value by today. A low-ℓ limit therefore constrains a *different combination* in the two models, so a bound quoted for one is not a bound for the other. The CDM column has a published precedent; the ΛCDM column is new.

Beyond v2.0.0. The published bounds are first order in the anisotropy parameter. The natural extension is the next order, and this project is placed to make it because **v1.5.0 supplies the $O(\varepsilon^2\ell)$ couplings a second-order argument needs** and no external code carries them. Stated as a direction and not a plan: it is gated on v1.5.0, and the correction's sign and size are unknown until it is done. It is recorded so the v2.0.0 table is built in a shape that can take it. See `provenance/ALMOST-EGS-BOUNDS-v1.md`.

11 Interpretive limits

This is an analytical reproducibility bundle for the associated paper and its supplement. It is not an empirical calibration, not a parameter-estimation pipeline, and not a Boltzmann code for general use. No new effect is claimed and none is reported.

Annals II's own §8.3.2 model is standard CDM — $h = 0.5$, $\Omega_B = 0.05$, flat and matter dominated — and the C_ℓ it prints there is a closed-form Sachs–Wolfe result on large scales with the transfer function set to unity. This release goes past that in one respect and one only: it computes the spectrum from the hierarchy rather than from the closed form, and it computes it for the ΛCDM model as well, which *Annals II* does not treat. The marginal cost is the expansion law and the late-time integrated term; every other ingredient — recombination, tight coupling, diffusion damping — is already required by the CDM case and is not additional. No new *physics* is introduced by Λ, and none is claimed.

What the release does **not** do is calibrate. Comparison is made against CMBFAST and against nothing else, over a derived range, as a check that the reconstruction is right — not as a fit. Parameter estimation and calibration against data are the **v2.0.0** deliverable.

Two statements must travel together, and neither stands alone: the nonlinear contribution $\delta\tau_{NL}$ **matters** — it is not negligible and not an artefact — and it **adds nothing**, redistributing power rather than creating it, because $\delta\tau_{NL} \propto \Phi\Phi$ makes its contribution to C_ℓ quartic in Φ while the linear C_ℓ is quadratic.